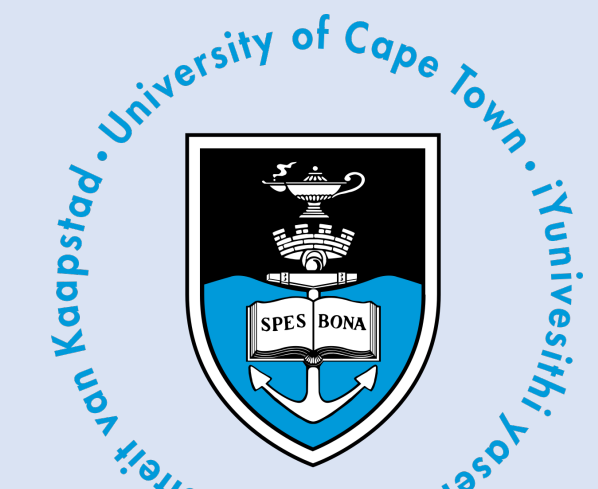


From Lead to Helium: Discovery Potential for Jet Quenching in the Smallest Collision Systems



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Based on CF, B. Bert, J. Brand, W. Vogelsang, and W. A. Horowitz, [arXiv:2512.17832](https://arxiv.org/abs/2512.17832)

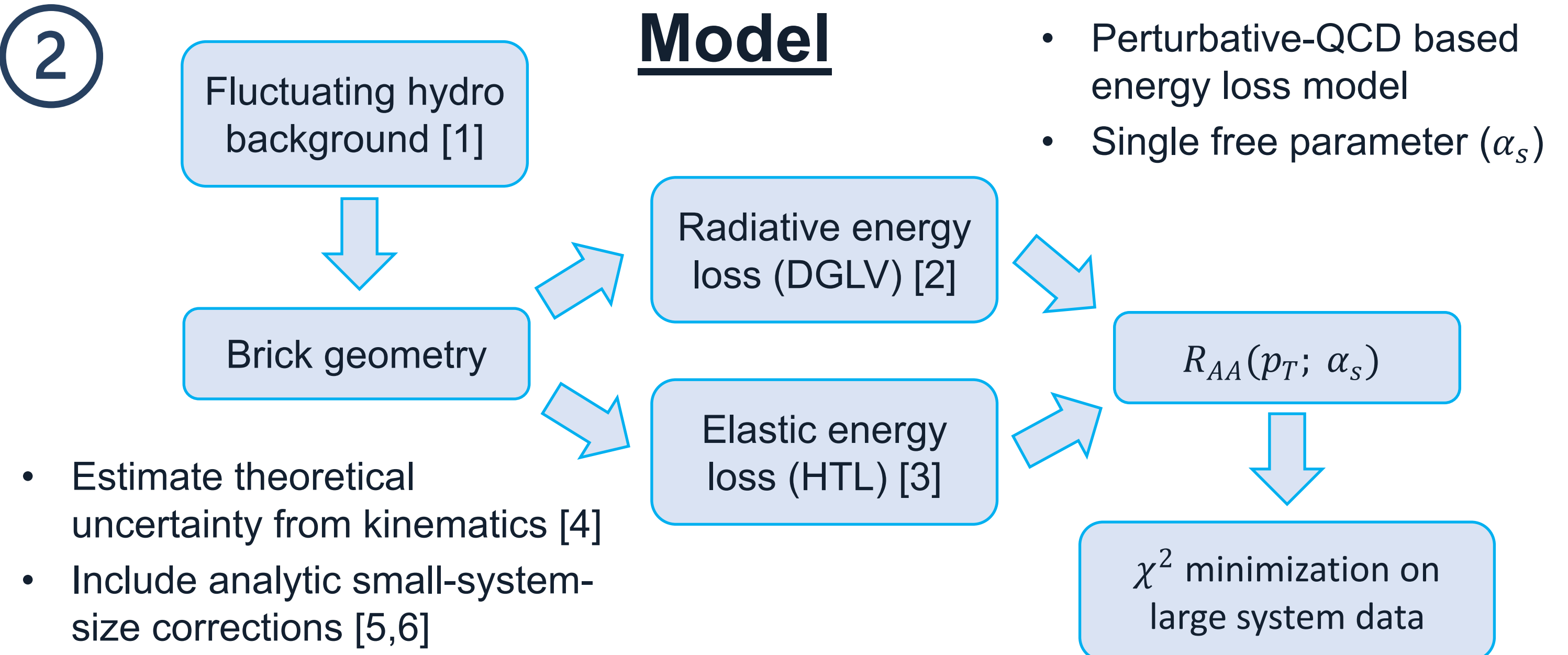
1

Introduction

Jet quenching has long been the missing link in establishing quark-gluon plasma (QGP) formation in small systems. While numerous other signatures of QGP formation have been observed in high-multiplicity pp and pA collisions, evidence for jet quenching has remained absent. Recently, O+O and Ne+Ne collisions at the LHC measured both elliptic flow and jet quenching, offering strong evidence for QGP formation in these small systems. An open question remains: how far can the heavy-ion paradigm be extended down in system size? That is, what is the smallest droplet of collective, flowing quark-gluon plasma? We present first predictions for suppression driven by final-state partonic energy loss and suppression driven solely by initial-state nPDF effects in a variety of extremely small, symmetric systems, including $^3\text{He} + ^3\text{He}$, $^4\text{He} + ^4\text{He}$, $^6\text{Li} + ^6\text{Li}$, and $^{10}\text{B} + ^{10}\text{B}$ collisions.

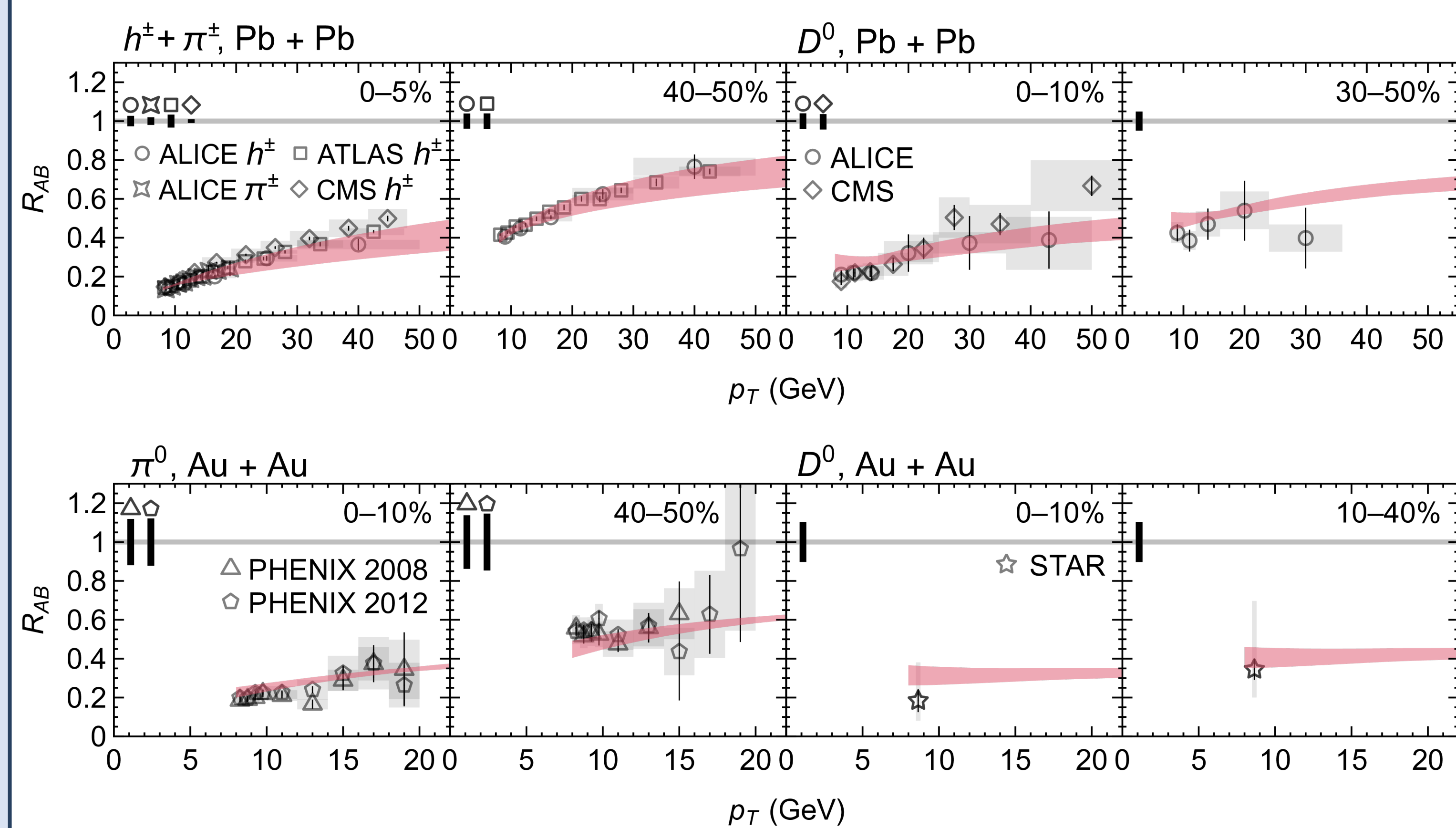
2

Model



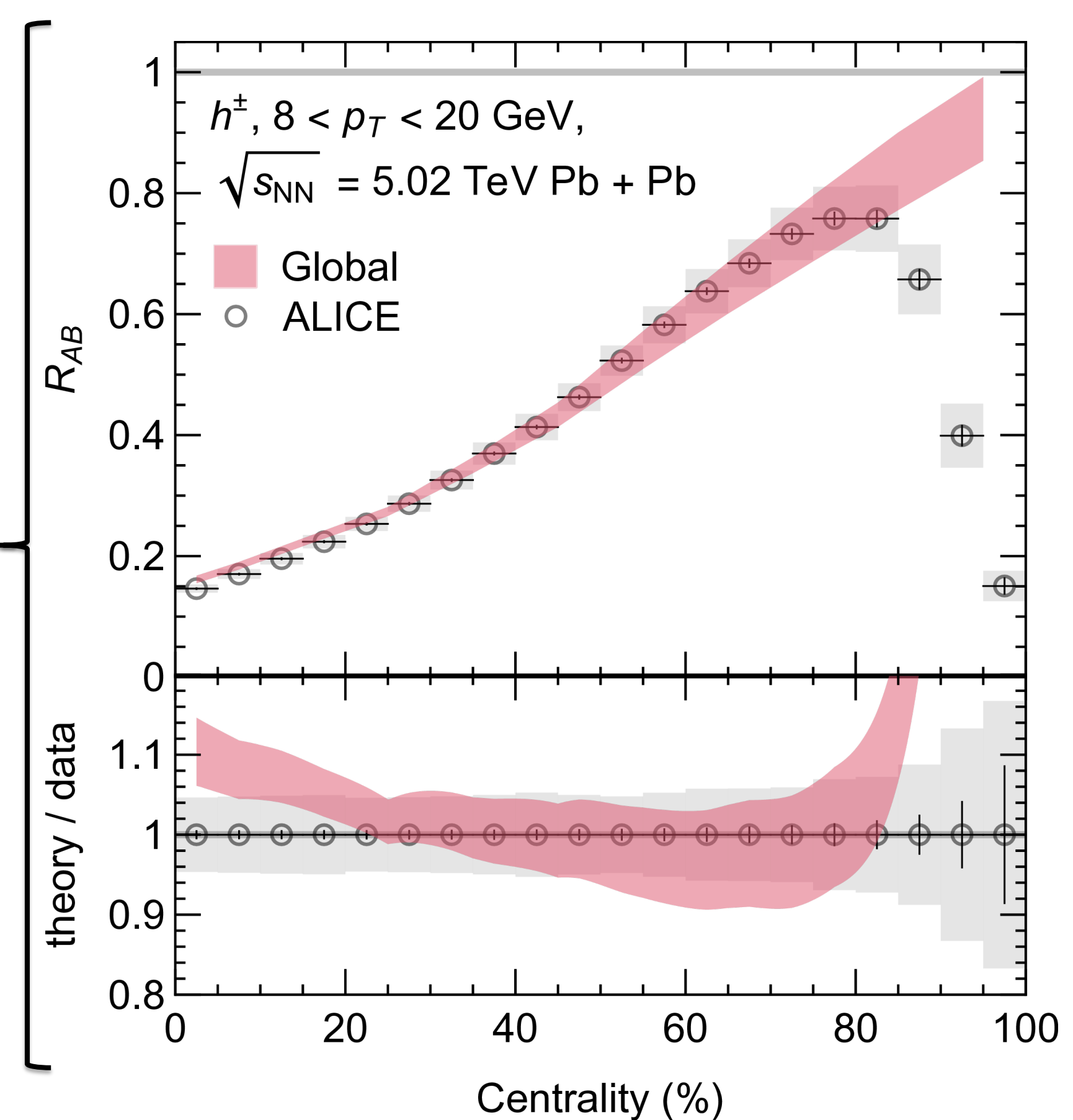
3

Calibration on Large Systems



- We calibrate the single model parameter α_s on high- p_T data from 0-50% centrality Pb + Pb and Au + Au collisions [4]
- The model provides a good description of the data across a wide range of p_T , $\sqrt{s}$, hadron species, and centrality

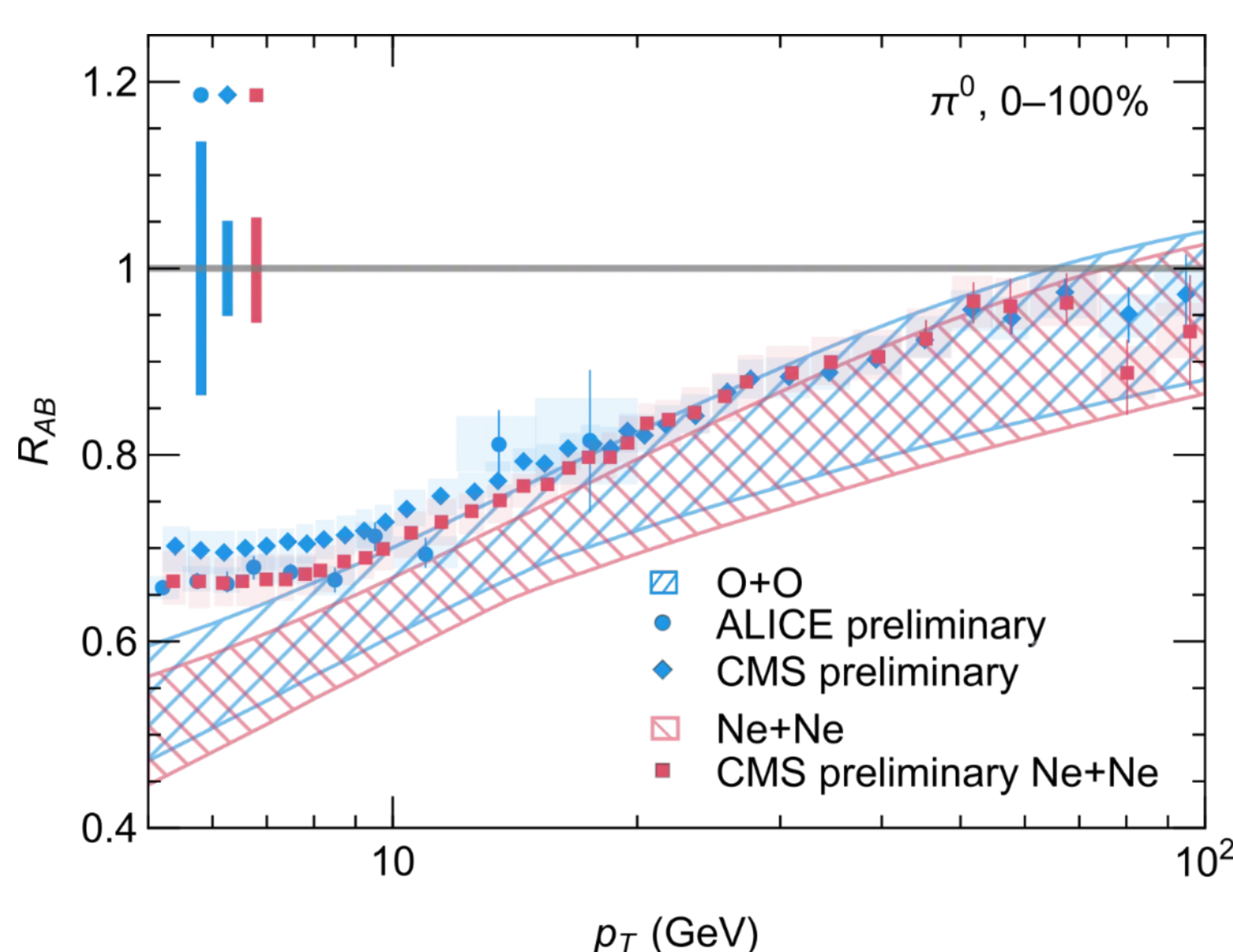
- In particular, the excellent description of the R_{AA} vs centrality motivates the ability of the model to be extrapolated to small systems



4

Light-ion collisions

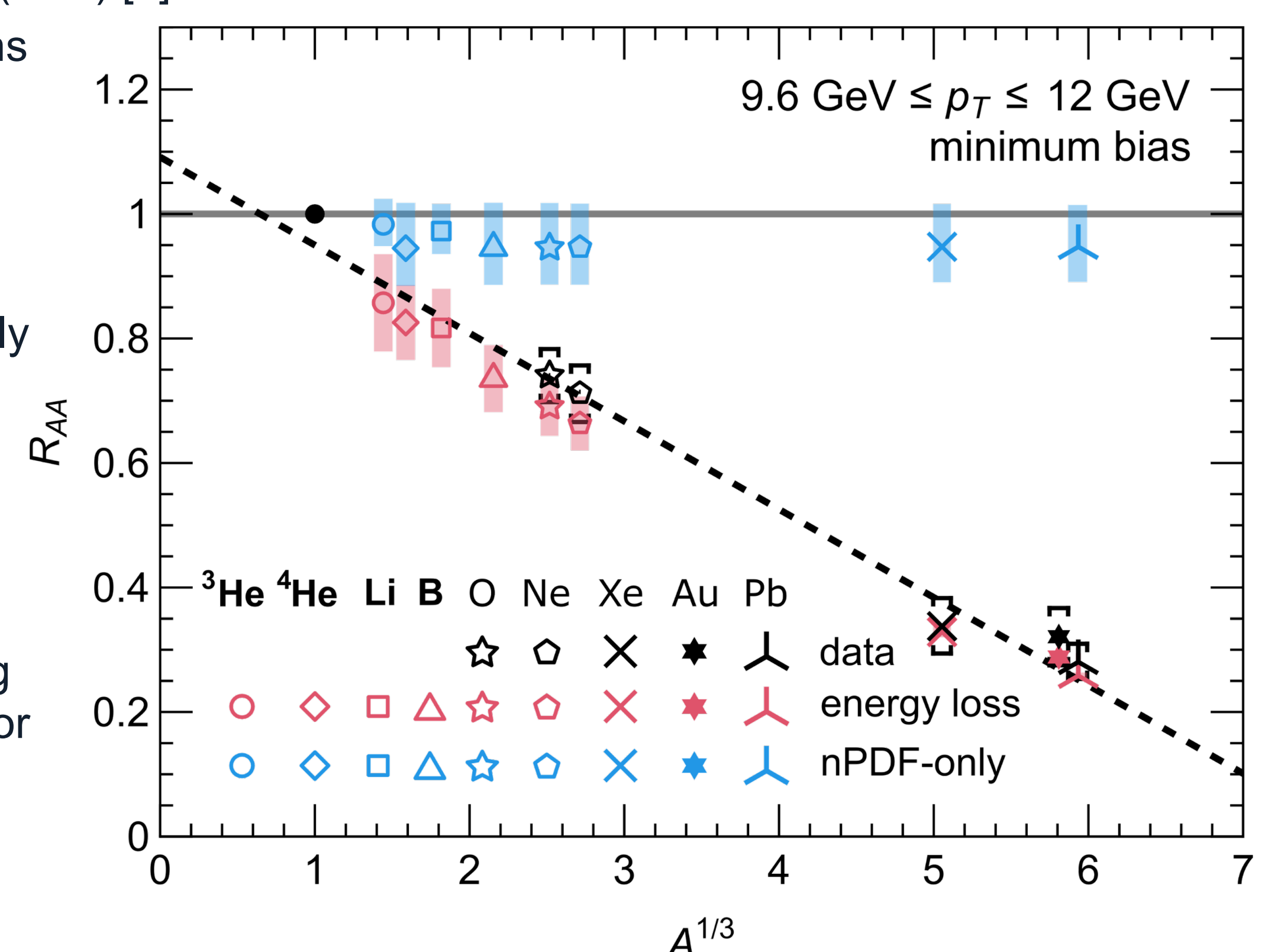
- Blind predictions for O+O and Ne+Ne suppression in excellent agreement with data [4]
- Slight overprediction of suppression at low p_T , likely due to missing medium-modification to hadronization in the model.



5

Even lighter-ion collisions

- We compare energy loss predictions (red) to available data (black) and nPDF-only baseline predictions (blue) [7]
- nPDF-only predictions are from NLO pQCD [8] with EPPS21 nPDF fit [9]
- We find approximately $R_{AA} \propto A^{1/3}$ for all systems
- ^3He and ^6Li have very small nPDF suppression and uncertainties, making them “sweet spots” for discovering energy loss in extremely small systems [7]



6

Future Work

- Theoretical framework that includes hard-soft correlations, relevant for centrality-selected small systems [10]
- Complementary studies
 - Hydrodynamic predictions for ^3He , ^4He , ^6Li , ^{10}B
 - Nuclear structure studies, e.g., alpha clustering; $^6\text{Li} \sim \alpha + d$, $^{10}\text{B} \sim 2\alpha + d$

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